1 Multilinear Forms

Let V be a vector space over F. For m a positive integer, let $V^m = \underbrace{V \times \cdots \times V}_{m \text{ times}}$ denote the m-fold Cartesian product of V with itself. That is, V^m is the set of all ordered m-tuples of elements of V.

Definition 1.1 (Multilinear form)**.** An m-linear form on V is a map $M : V^m \to F$ which is linear in each entry when all other entries are held fixed. That is, for each $k \in \{1, \ldots, m\}$ and all $u_1, \ldots, u_m \in V$, the map

$$v \mapsto M(u_1, \ldots, u_{k-1}, v, u_{k+1}, \ldots, u_m)$$

is an element of V^*.

The set of m-linear forms on V is denoted by $V^{(m)}$. An element of $V^{(m)}$ for some m is called a *multilinear form*. A 1-linear form is the same as an element of V^*. A 2-linear form is the same as a bilinear form.

By a proof analogous to the one for $V^{(2)}$, one can prove that $V^{(m)}$ is a vector space for every positive integer m with respect to the usual operations

$$\begin{aligned}(M_1 + M_2)(v_1, \ldots, v_m) &:= M_1(v_1, \ldots, v_m) + M_2(v_1, \ldots, v_m)\\ (\alpha M)(v_1, \ldots, v_m) &:= \alpha M(v_1, \ldots, v_m).\end{aligned}$$

Exercise 1.2. Suppose $B_1, B_2 \in V^{(2)}$. Define $M : V^4 \to F$ by $M(v_1, v_2, v_3, v_4) = B_1(v_1, v_2)B_2(v_3, v_4)$. Show that $B_4 \in V^{(4)}$.

Exercise 1.3. Define $M : (\mathscr{L}(V))^m \to F$ by $M(T_1, \ldots, T_m) = \operatorname{tr}(T_1 \cdots T_m)$. Show that $M \in V^{(m)}(\mathscr{L}(V))$.

2 Alternating Multilinear Forms

Definition 2.1 (Alternating multilinear form)**.** An m-linear form M is *alternating* if

$$M(v_1,\ldots,v_i,\ldots,v_j,\ldots,v_m) = -M(v_1,\ldots,v_j,\ldots,v_i,\ldots,v_m)$$

whenever we interchange two vectors v_i and v_j. We denote the set of alternating m-linear forms by $V_{\text{alt}}^{(m)}$. Since a 1-linear map is trivially alternating, $V^{(1)} = V^*$. An alternating 2-linear form is the same as an alternating bilinear form.

The following is an equivalent definition of an alternating m-linear form.

Proposition 2.2. *$M \in V^{(m)}$ is alternating if and only if $M(v_1,\ldots,v_m) = 0$ whenever $v_i = v_j$ for two distinct values of j and k in $\{1,\ldots,m\}$.*

Exercise 2.3. Prove Proposition 2.2.

Proposition 2.4. *If $M \in V_{alt}^{(m)}$ and $\{v_1,\ldots,v_m\}$ is linearly dependent, then $M(v_1,\ldots,v_m) = 0$.*

Exercise 2.5. Prove Proposition 2.4.

Corollary 2.6. *If $m > \dim V$, then $\dim V_{alt}^{(m)} = 0$.*

Exercise 2.7. Prove Corollary 2.6.

3 Permutations

Here we summarize some facts about permutations.

- A *permutation* of the set $A = \{1, 2, \ldots, m\}$ is a bijection $\sigma : A \to A$, which can be thought of as a reordering of the list $1, 2, \ldots, m$ to a new order $\sigma(1), \sigma(2), \ldots, \sigma(m)$. A convenient way to describe a permutation σ is by a matrix

$$\begin{bmatrix} 1 & 2 & \cdots & m \\ \sigma(1) & \sigma(2) & \cdots & \sigma(m) \end{bmatrix}$$

  where the inputs form the first row and the corresponding outputs form the second row. For example the permutation $1 \mapsto 3$, $2 \mapsto 1$, $3 \mapsto 2$ is represented by the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}$$

  Note that nothing is multiplied by this matrix; it just keeps track of the inputs/outputs.

- Permutations of A form a *group* under composition (which is an associative operation), denoted by S_m. [1] For $\sigma, \tau \in S_m$, we write $\tau\sigma$ for the composition $\tau \circ \sigma$. The identity element of the group is the trivial permutation $\begin{bmatrix} 1 & 2 & \cdots & m \\ 1 & 2 & \cdots & m \end{bmatrix}$. We denote the inverse of a permutation σ by σ^{-1}.

- A *cyclic permutation* or *r-cycle*, denoted $(a_1\ a_2\ \cdots\ a_r)$, where the a_i are distinct, is the permutation σ such that $\sigma(a_i) = a_{i+1}$ for $i = 1, \ldots, r$, $\sigma(a_r) = a_1$, and $\sigma(a_\ell) = a_\ell$ for $\ell > r$. A 2-cycle is also called a *transposition*.

- Cycles $(a_1\ \cdots\ a_r)$ and $(b_1\ \cdots\ b_s)$ are *disjoint* if the sets $\{a_1, \ldots, a_r\}$ and $\{b_1, \ldots, b_r\}$ have no elements in common.

- Any permutation σ can be written as a product of disjoint transpositions. We say that σ is *even* or *odd* depending on whether it is the product of an even or odd number of transpositions.

- We define the *sign* of a permutation σ, denoted by $\operatorname{sgn}(\sigma)$, to be $+1$ or -1 depending on whether the permutation is even or odd. This satisfies $\operatorname{sgn}(\tau\sigma) = \operatorname{sgn}(\tau)\operatorname{sgn}(\sigma)$.

[1] A *group* is a set G with an associative binary operation, such that G has an identity element, and every element of G has an inverse.

Example 3.1. Consider the permutation

$$\sigma = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 1 & 3 \end{bmatrix} \in S_5.$$

To write σ as a product of disjoint cycles, start with any element of $\{1,2,3,4,5\}$, say 1, and apply σ to it repeatedly until we return to the initial element; this gives a cycle: $1 \mapsto 2 \mapsto 4 \mapsto 1$, or (124). Next, repeat this procedure beginning with any of the remaining elements, say 3, to get a second cycle: $3 \mapsto 5 \mapsto 3$, or (35). Since all elements of $\{1,2,3,4,5\}$ are now accounted for, $\sigma = (124)(35)$. Noting that the product $(14)(12)$ also sends $1 \mapsto 2 \mapsto 4 \mapsto 1$, we have $(124) = (14)(12)$, and therefore $\sigma = (14)(12)(35)$, so we see that $\operatorname{sgn}(\sigma) = -1$.

Example 3.2. More generally, the decomposition

$$(a_1 a_2 \cdots a_r) = (a_1 a_r)(a_1 a_{r-1}) \cdots (a_1 a_3)(a_1 a_2).$$

shows that an r-cycle is an even permutation if and only if r is odd, and an odd permutation if and only if r is even. Thus, a faster way to compute the sign of a permutation is to decompose it into a product of cycles and to count the number of cycles of even length. For example, the permutation $\sigma = (124)(35)$ in Example 3.1 is odd because (124) is even and (35) is odd.

4 Permutation Action on Multilinear Forms

Definition 4.1 (Group action)**.** Let G be a group and X be a set. A *left group action* is a mapping

$$\begin{aligned} G \times X &\to X \\ (g,x) &\mapsto gx \end{aligned}$$

which satisfies the following two properties:

(i) $ex = x$, where e is the identity element of G and $x \in X$.

(ii) $h(gx) = (hg)x$ for all $g,h \in G$ and $x \in X$.

A right group action is defined similarly.

Definition 4.2 (Orbit)**.** If there is a left action of G on X, then the set

$$\mathcal{O}_x := \{gx \in X : g \in G\}$$

is called the *orbit* of the element $x \in X$.

We can define a left action of S_m on $V^{(m)}$ as

$$\begin{aligned} S_m \times V^{(m)} &\to V^{(m)} \\ (\sigma, M) &= \sigma M, \end{aligned}$$

where

$$(\sigma M)(v_1, \ldots, v_k) = M(v_{\sigma(1)}, \ldots, v_{\sigma(m)}).$$

Note this is indeed a left action of S_m on $V^{(m)}$, since

(i) If σ is the trivial permutation, then

$$(\sigma M)(v_1,\ldots,v_m) = M(v_{\sigma(1)},\ldots,v_{\sigma(m)}) = M(v_1,\ldots,v_m),$$

hence $\sigma M = M$.

(ii) If $\sigma,\tau \in S_m$, then

$$\begin{aligned}
\tau(\sigma M)(v_1,\ldots,v_m) &= (\sigma f)(v_{\tau(1)},\ldots,v_{\tau(m)}) \\
&= (\sigma M)(w_1,\ldots,w_m) \qquad (\text{letting } w_i = v_{\tau(i)}) \\
&= M(w_{\sigma(1)},\ldots,w_{\sigma(m)}) \\
&= m(v_{\tau(\sigma(1))},\ldots,v_{\tau(\sigma(m))}) \\
&= M(v_{\tau\sigma(1)},\ldots,v_{\tau\sigma(m)}) \\
&= (\tau\sigma)M(v_1,\ldots,v_m),
\end{aligned}$$

hence $\tau(\sigma M) = (\tau\sigma M)$.

Proposition 4.3. *$M \in V^{(m)}$ is alternating if and only if $\sigma M = sgn(\sigma)M$ for all $\sigma \in S_m$.*

Proof. If $\sigma M = \operatorname{sgn}(\sigma)M$ for each $\sigma \in S_m$, then in particular, when σ is a transposition we have $\sigma M = \operatorname{sgn}(\sigma)M = -M$. Hence, M is alternating.

Suppose now that M is alternating and let $\sigma \in S_m$. As noted in the previous section, σ can be written as a product of disjoint transpositions. Since M picks up a negative sign from each transposition, we will have $\sigma M = M$ if σ is even and $\sigma M = -M$ if σ is odd. Hence, $\sigma M = \operatorname{sgn}(\sigma)M$. □

Proposition 4.4. *Let V be an n-dimensional vector space and let $\{e_1,\ldots,e_n\}$ be a basis of V. For any $v_1,\ldots,v_n \in V$ there exist scalars b_{jk} such that $v_k = \sum_{j=1}^{n} b_{jk}e_j$.*

If $M \in V^{(n)}$, then

$$M(v_1,\ldots,v_n) = M(e_1,\ldots,e_n)\sum_{\sigma\in S_n} sgn(\sigma)b_{\sigma(1)1}\cdots b_{\sigma(n)n}$$